董媛 DONG YUAN

MSc AI in Medicine| Biomedical Engineering | Healthcare AI

Phone: +65 9051 2151 Email: yuan027@e.nut.edu.sg Languages: Chinese (Mandarin, Native), English (IELTS 7)

Education

Nanyang Technological University (NTU) · MSc AI in Medicine Jul 2026 — Aug 2027 (Expected)

Relevant Coursework: Machine Learning, Deep Learning, NLP & Large Language Models, Multimodal Learning

Shandong University · BEng in Biomedical Engineering Sep 2020 — Jun 2026

Relevant Coursework: Digital Signal Processing, Digital Image Processing, Signals & Systems, Control Theory

Research / Capstone Project

Wearable PPG Motion Artifact Suppression 2025.12 — 2026.05

BEng Capstone / Research Project | Shandong University

- Analyzed time- and frequency-domain characteristics of PPG and acceleration signals and designed an adaptive filtering approach for motion-artifact suppression in wearable heart-rate monitoring.
- Implemented LMS/NLMS adaptive filtering and explored a variable-step-size VS-LMS improvement for slow walking, jogging and running scenarios.
- Evaluated 187 multi-wavelength PPG recordings from 15 subjects (530/660/960 nm, 25 Hz) using SNR, heart-rate error (HRE) and PCC; jogging SNR improved by 4.67 dB and HRE decreased from 6.18 to 1.29 bpm (~79%).
- Used MATLAB and Python for signal preprocessing, algorithm implementation, batch experiments and result visualization.

TongueDX — Healthcare Innovation & Entrepreneurship Project 2026.08 — Present

Nanyang Technological University

- Initiated the TongueDX concept and helped frame a potential AI-enabled healthcare solution around an identified healthcare problem.
- Developing a business plan covering the proposed value proposition, target users, market opportunity and commercialization considerations.
- Collaborating with a multidisciplinary team to assess feasibility and potential pathways toward healthcare implementation.

Smartphone-Based Physiological Signal Processing 2025.09 — 2025.12

ShanDong University

- Studied PPG and respiratory signal mechanisms and waveform characteristics and implemented signal denoising and demodulation workflows.
- Validated heart-rate and blood-oxygen measurement algorithms using real measurements, analyzed error sources and performed comparative calibration.
- Developed a mobile interface to present algorithm outputs and evaluate heart-rate and respiratory-rate estimation.

ECG Signal Processing & Feature Analysis 2025.02 — 2025.06

ShanDong University

- Built an ECG processing pipeline on PhysioNet 2017 single-lead ECG data, including filtering, R-peak detection and HRV analysis in MATLAB.
- Combined band-pass filtering, Mexican Hat wavelet local-maximum search, dynamic thresholds and minimum RR-interval constraints for robust R-peak detection under noise.
- Computed HR and SDNN from RR-interval sequences to support feature-based screening of abnormal cardiac rhythms.

Technical Skills

- **Programming:** C/C++, Python, MATLAB
- **Biomedical Signal Processing:** Adaptive filtering, digital filtering, time-frequency analysis, sensor signal denoising, ECG/PPG
- **AI / Machine Learning:** Machine Learning, Deep Learning, Medical AI, Medical Imaging, Multimodal Learning, NLP & LLMs
- **Hardware & Tools:** Keil, Multisim, analog front-end design, amplifier and filter design, oscilloscope, multimeter